Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Sinha, Shantanu Ph.D.		
Project Title:	In vivo Stress-Strain Dynamics in Human Muscle		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ssinha@mednet.ucla.edu	**Fax:**	FY 310-794-6613
PI Organization Type:	UNIVERSITY	**Phone:**	310-825-2320
Organization Name:	University of California, Los Angeles		
PI Address 1:	BL-428 UCLA School of Medicine		
PI Address 2:	10833 LeConte Avenue		
PI Web Page:			
City:	Los Angeles	**State:**	CA
Zip Code:	90095-1721	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	08/01/2001	**End Date:**	10/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	4
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edgerton, Reggie (University of California, Los Angeles) Elashoff, Robert (University of California, Los Angeles) Hodgson, John (University of California, Los Angeles) Lai, Alex (University of California, Los Angeles) Roy, Roland (University of California, Los Angeles)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Muscle atrophy refers to the wasting or loss of muscle tissue resulting from lack of activity due to several reasons, including prolonged bed rest, limb suspension and immobilization due to disease and trauma, aging and spaceflight. Regarding space mission in microgravity environment, muscle atrophy and the accompanying neuromuscular disorders have been recognized as serious obstacles to be overcome for long-duration space exploration. Although many research projects have well documented morphological, biological and physiological changes in atrophied muscles using spaceflight and ground-based experimental models, it is virtually unknown what functional implications the changes in mechanical properties and structure of intramuscular tendinous tissues have. In addition, in order to investigate this issue in human muscles in vivo under the environment closer to physiological condition, it is required to develop a non-invasive measurement technique. For the last three years, our laboratory has developed, verified, and perfected a completely non-invasive measurement technique using velocity-encoded, phase contrast MR imaging combined with a custom-build MR-compatible muscle strength testing apparatus. This method enabled us to accurately quantify muscle volume and strength and strain dynamics of muscular-tendinous structures in human lower leg. Using this method, we examined the effect of a period of decreased mechanical load/suspension and the changes which occur upon recovery of the muscle to normal loading condition. For 13 subjects, atrophy was induced using a 4-week unilateral limb suspension (ULLS) model followed by 6-week rehabilitation. The ULLS model used was very effective for inducing localized atrophy, completely reversible and well-tolerated by the normal volunteers. Preliminary results of the previous ULLS study (performed in the prior years of the grant) showed that the decrease in muscle volumes following ULLS was proportionate to similar changes in relative strain distribution of the tendinous structures, suggesting that the structure and function are tightly coupled. Moreover, preliminary results of the most recent ULLS study (performed in 2004) demonstrated that absolute strain distribution (assessed with 10 and 20 % Pre-ULLS MVC) of tendinous structures changed following ULLS, possibly due to different motor unit recruitment strategy and altered mechanical properties of the structures. In addition, a different rehabilitation paradigm with moderate resistance training protocol (using loading condition less than subject's body weight) was tried out and found to be very effective. For the coming years, we would like to pursue better understanding of functional implications of architecture and mechanical linkages in atrophied muscles using MR Tractography on the top of current method. We strongly believe that future research study will provide valuable information to understand the structure and functional properties of normal and atrophied muscle and to develop more effective exercise regimen for astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The method developed herein, namely, using a non-invasive method of monitoring structural and functional differences between normal and atrophied muscles has a tremendous potential for Earth-based clinical use. Atrophy can result not only from micro-gravity as in the case of astronauts in space flights, but also from prolonged bed-rest or immobilization from casting. In these cases, the extent to which the muscle normalizes subsequent to rehabilitation is a very important clinical question, and can be addressed quantitatively by the method developed herein.
Task Progress:	During the project year 2005, we interviewed and recruited 6 additional ULLS subjects after IRB approval. With one withdrawn due to personal reasons, 5 remaining subjects (4 males and 1 female) participated and successfully completed all aspects of the study including 4-week limb suspension, 6 MR sessions, and 6-week physical rehabilitation program. This study fulfilled 3 important objectives: 1. It greatly improved the statistical basis of our previously published findings of atrophy-induced changes in muscle volume & strength, the contraction velocity, and strain dynamics of the aponeurosis-tendon complex. As before, our method could detect differential changes in different muscle groups, as well as track the return to normalcy as the subjects were given physical rehabilitation/therapy. 2. We investigated the effect of atrophy on the Achilles tendon compliance. This is a new parameter that has not been studied previously in this project. Our methodology accurately quantified the tendon stiffness (N/m) in all subjects and proved its efficacy in monitoring stiffness changes after onset of atrophy. Our results show statistically significant decrease in stiffness immediately after 4 weeks of chronic unloading, then gradual increase toward the pre-atrophy level during the 6-week physical rehabilitation program. This finding clearly demonstrates fast physiological response of the muscle-tendon complex toward homeostasis. It also seems to indicate that 6 weeks of physical rehabilitation program is fairly sufficient in restoring the mechanical properties of the tendon to its normal state. 3. We calculated the spatial shear strain distribution along the proximal region of the aponeurosis-tendon complex (above MG insertion). This also is a new parameter previously not presented in this project. Similar to the distal portion of the complex, the proximal region exhibited heterogeneous strain distribution during isometric contraction.
Bibliography Type:	Description: (Last Updated: 02/02/2007)
Articles in Peer-reviewed Journals	Finni T, Hodgson JA, Lai AM, Edgerton VR, Sinha S. "Mapping of movement in the isometrically contracting human soleus muscle reveals details of its structural and functional complexity." J Appl Physiol. 2003 Nov;95(5):2128-33. Epub 2003 Jul 11. PMID: 12857769 , Nov-2003
Articles in Peer-reviewed Journals	Finni T, Hodgson JA, Lai AM, Edgerton VR, Sinha S. "Nonuniform strain of human soleus aponeurosis-tendon complex during submaximal voluntary contractions in vivo." J Appl Physiol. 2003 Aug;95(2):829-37. Epub 2003 Apr 25. PMID: 12716873 , Aug-2003
Articles in Peer-reviewed Journals	Finni T, Hodgson JA, Lai AM, Edgerton VR, Sinha S. "Muscle synergism during isometric plantarflexion in achilles tendon rupture patients and in normal subjects revealed by velocity-encoded cine phase-contrast" Clin Biomech (Bristol, Avon). 2006 Jan;21(1):67-74. Epub 2005 Sep 27. PMID: 16194588 , Jan-2006
Articles in Peer-reviewed Journals	Hodgson JA, Finni T, Lai AM, Edgerton VR, Sinha S. "Influence of structure on the tissue dynamics of the human soleus muscle observed in MRI studies during isometric contractions." J Morphol. 2006 May;267(5):584-601. PMID: 16453292 , May-2006
Articles in Peer-reviewed Journals	Lee HD, Finni T, Hodgson JA, Lai AM, Edgerton VR, Sinha S. "Soleus aponeurosis strain distribution following chronic unloading in humans: an in vivo MR phase-contrast study." J Appl Physiol. 2006 Jun;100(6):2004-11. Epub 2006 Jan 19. PMID: 16424072 , Jun-2006

Articles in Peer-reviewed Journals	Sinha S, Hodgson JA, Finni T, Lai AM, Grinstead J, Edgerton VR. "Muscle kinematics during isometric contraction: development of phase contrast and spin tag techniques to study healthy and atrophied muscles." J Magn Reson Imaging. 2004 Dec;20(6):1008-19. PMID: 15558560 , Dec-2004